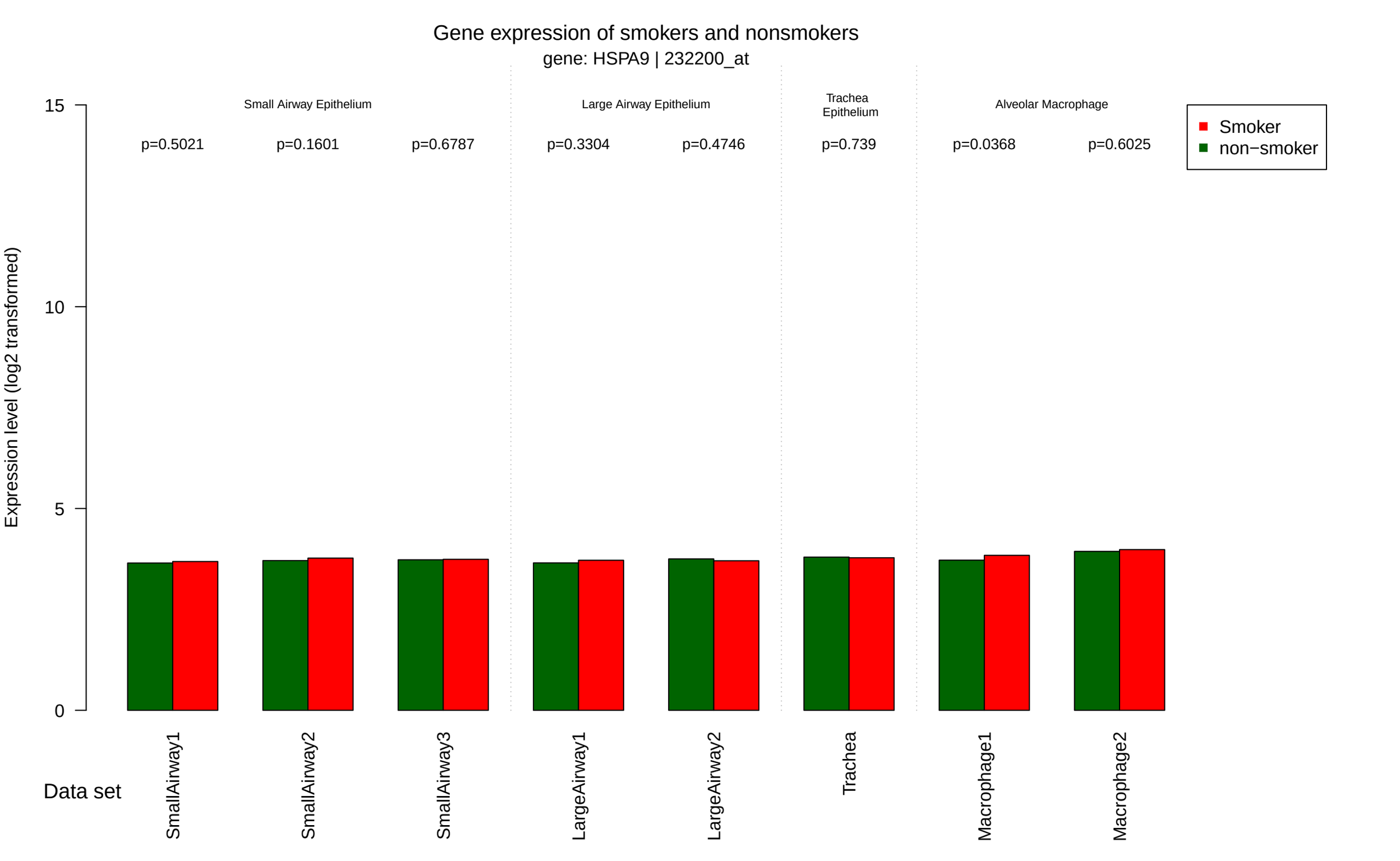
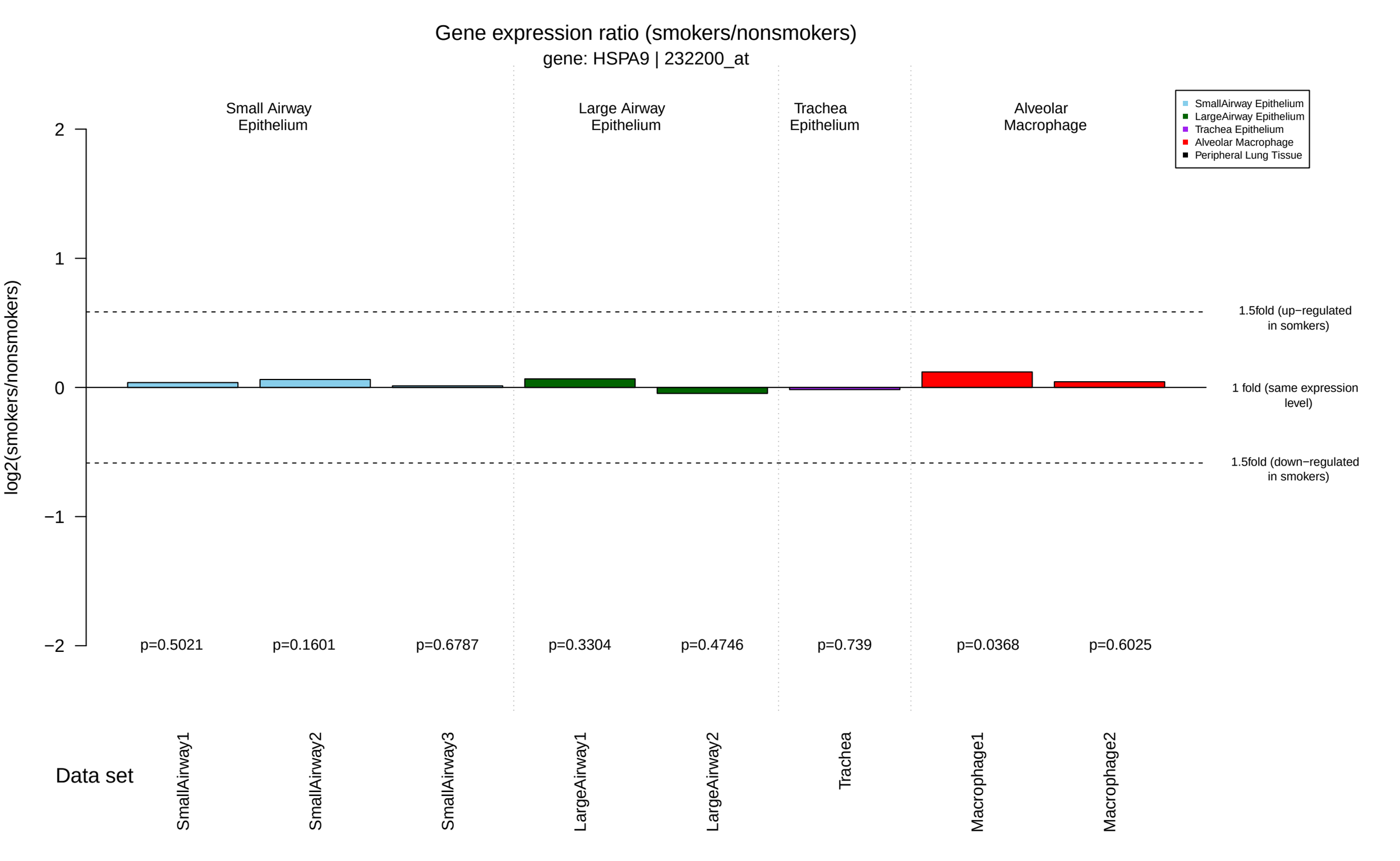
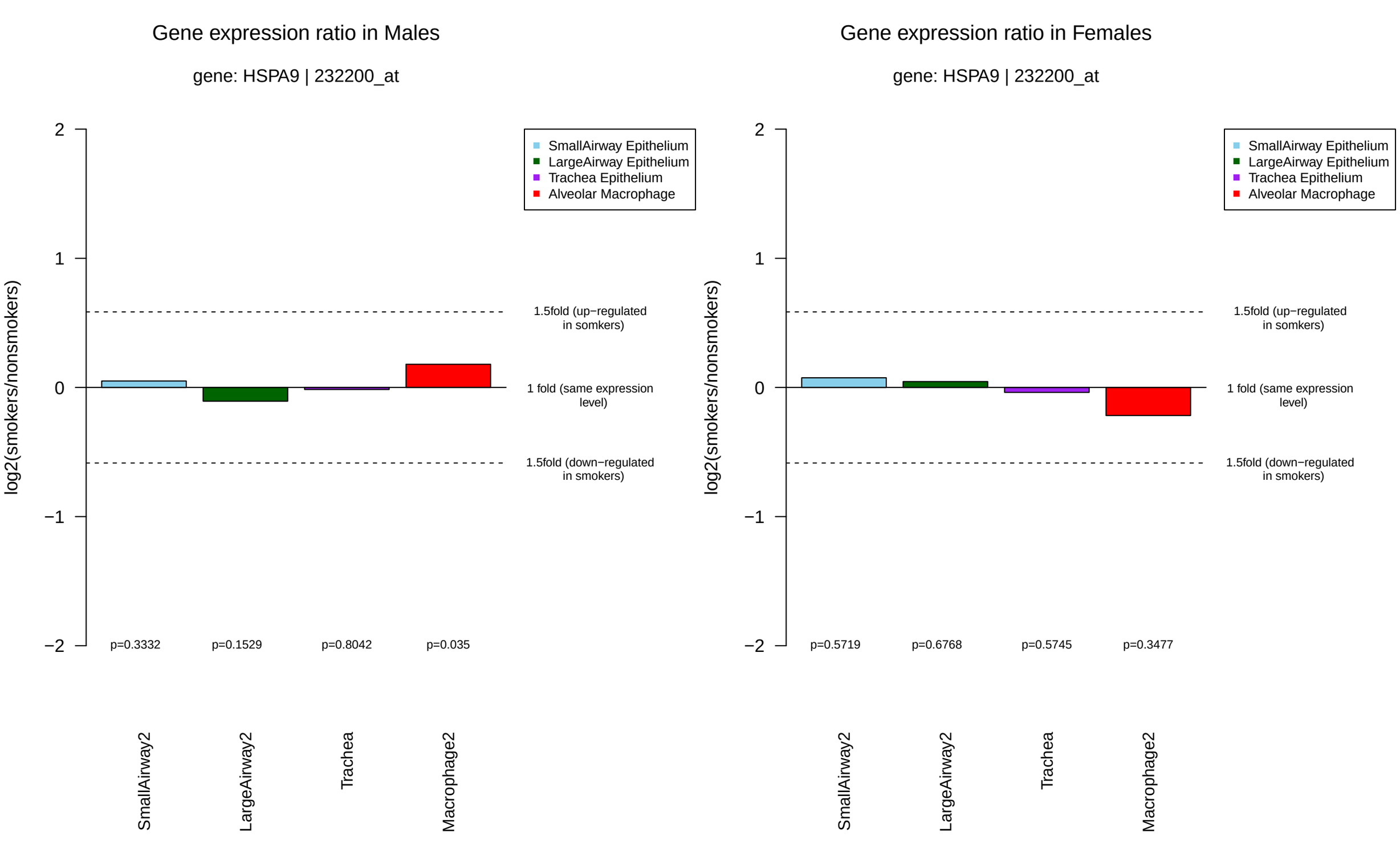
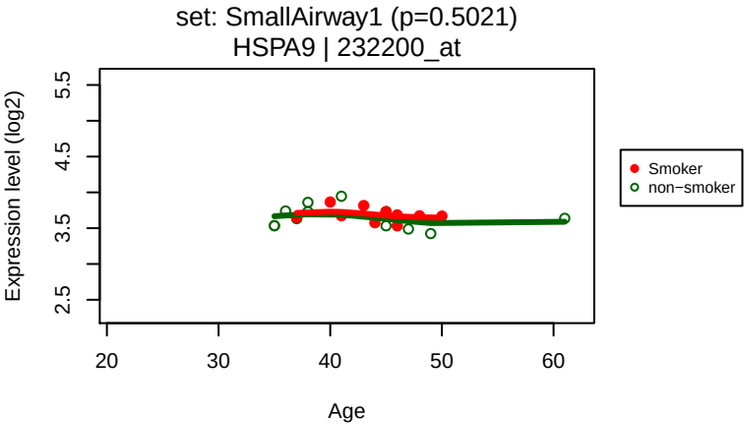
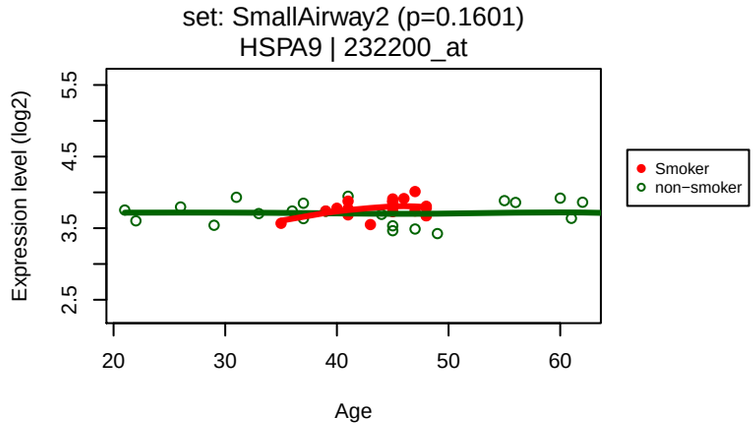
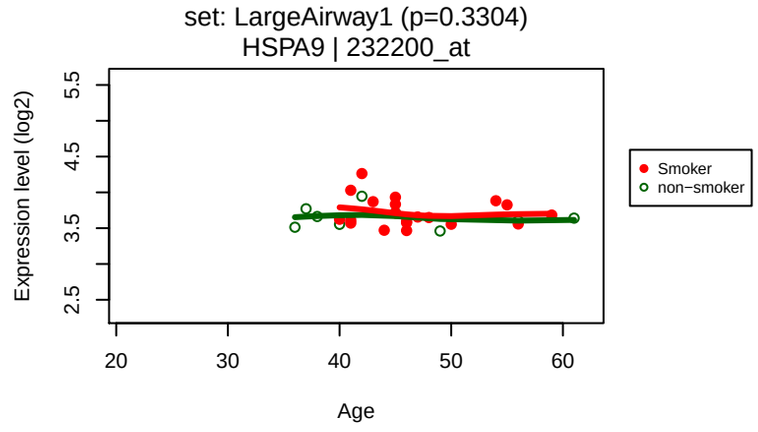

Gene expression of smokers and nonsmokers
gene: HSPA9 | 232200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5021
p=0.1601
p=0.6787
p=0.3304
p=0.4746
p=0.739
p=0.0368
p=0.6025
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

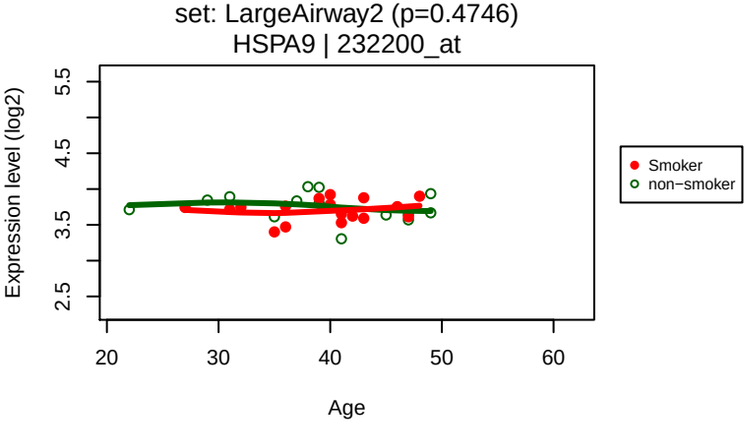

Gene expression ratio (smokers/nonsmokers)
gene: HSPA9 | 232200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5021
p=0.1601
p=0.6787
p=0.3304
p=0.4746
p=0.739
p=0.0368
p=0.6025
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

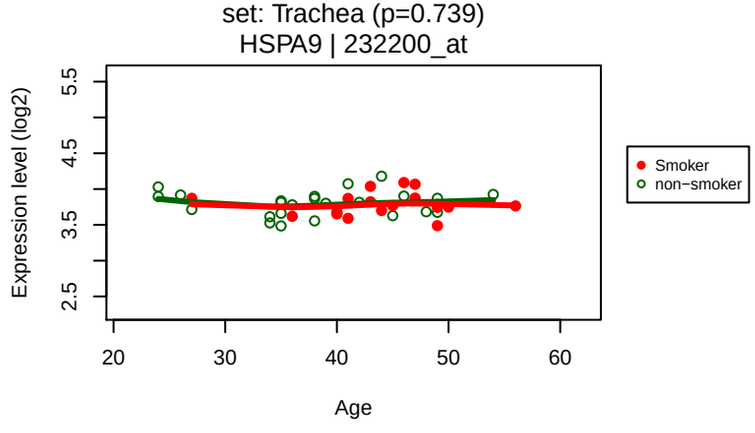

Gene expression ratio in Males
gene: HSPA9 | 232200_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3332
p=0.1529
p=0.8042
p=0.035
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA9 | 232200_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5719
p=0.6768
p=0.5745
p=0.3477
SmallAirway2
LargeAirway2
Trachea
Macrophage2

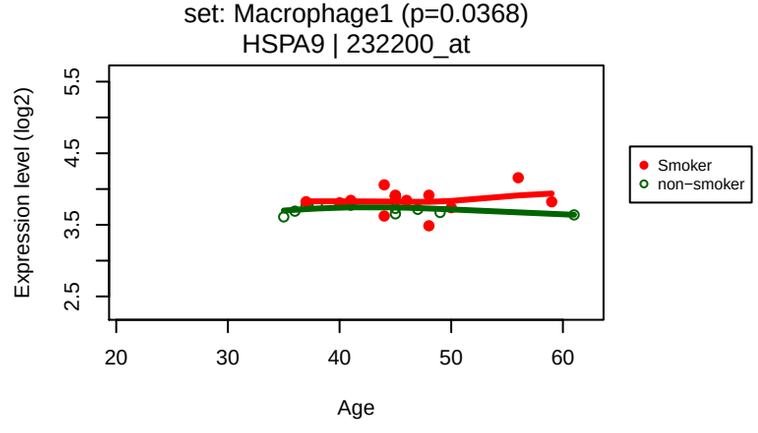

set: SmallAirway1 (p=0.5021)
HSPA9 | 232200_at
Expression level (log2)
Age
Smoker
non−smoker

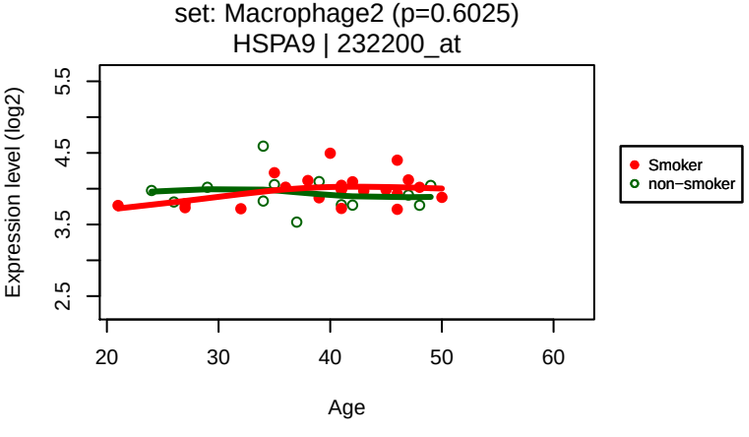

set: SmallAirway2 (p=0.1601)
HSPA9 | 232200_at
Expression level (log2)
Age
Smoker
non−smoker

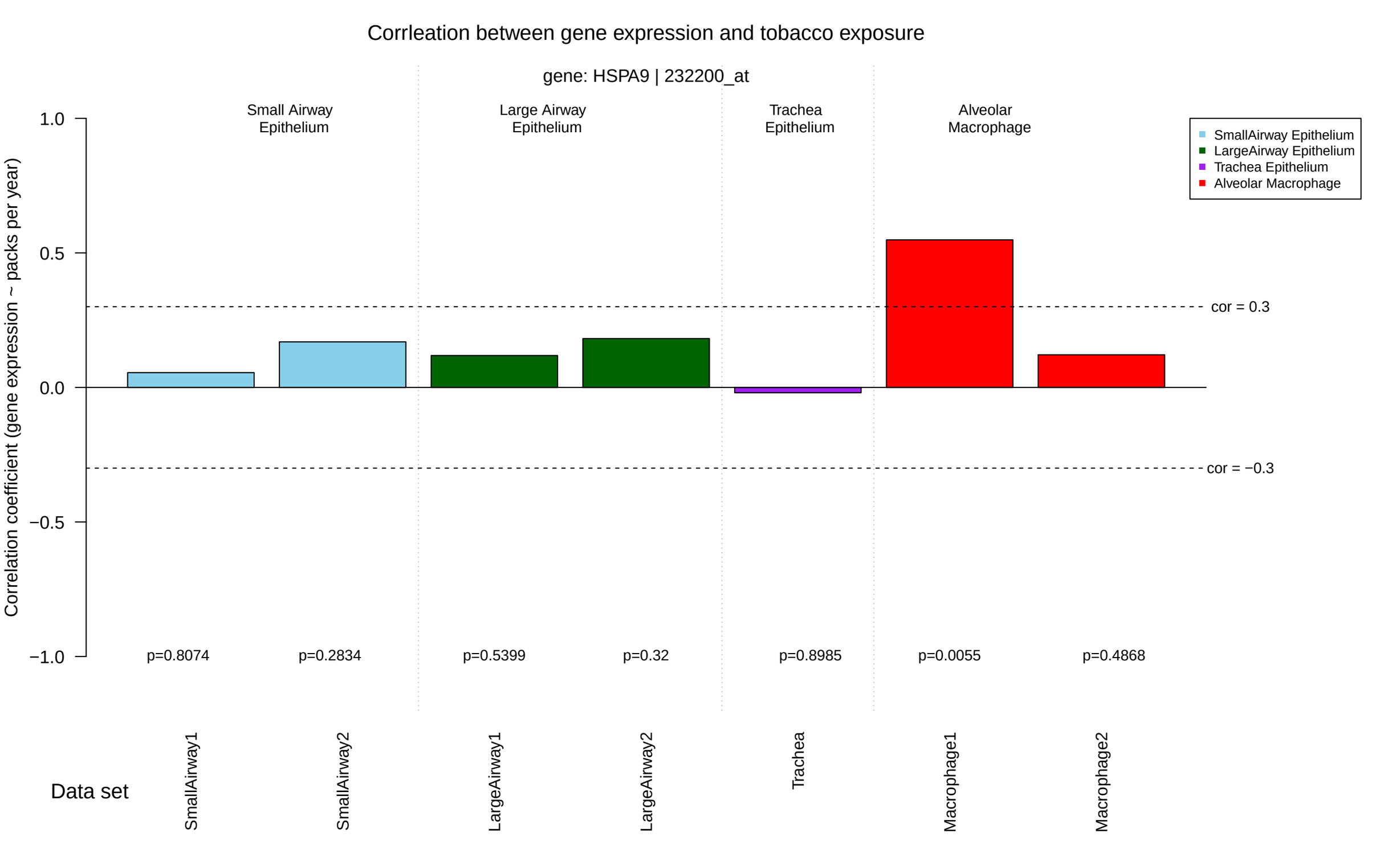

set: LargeAirway1 (p=0.3304)
HSPA9 | 232200_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4746)
HSPA9 | 232200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.739)
HSPA9 | 232200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0368)
HSPA9 | 232200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6025)
HSPA9 | 232200_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA9 | 232200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8074
p=0.2834
p=0.5399
p=0.32
p=0.8985
p=0.0055
p=0.4868
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2